



Endless storms on Jupiter

NASA's Juno mission to Jupiter has used its JunoCam to reveal large, 'endless storms' at the poles of the planet. <https://digit.in/jan19-73>



Toxic moon dust

Studies reveal that minerals in lunar dust react with human cells; cause respiratory problems including lung cancer. <https://digit.in/jan19-74>

actually be able to see, in the VR world, the action they want to perform with their amputated limb, which reduces the pain considerably. According to a recent study conducted in Sweden, about half of the participants reported a 50 per cent reduction in pain after going through VR treatment for their phantom limb pain.

Kim Bullock, the founder and director of Stanford's Neurobehavioural and Virtual Reality clinics and laboratories states that VR therapy assists in tricking our 'reptilian brain' which utilises pain as a mechanism to make us avoid using injured parts of our body.

Virtual reality also effectively enhances forms of distraction used previously to reduce pain such as watching television, listening to music or, in the case of children, playing with toys. Pain requires you to pay attention to actually feel it, and distractions reduce the intensity of the pain felt. These archaic treatments have been extensively studied and have proven to reduce pain and stress in patients suffering from pain. Virtual reality ups the ante by providing complete immersion, by consuming multiple senses, therefore providing the ultimate form of distraction. It doesn't just end there, VR treatments also train patients into learning deep breathing exercises, helping them with biofeedback and controlling their heartbeat and brain waves through VR activities, according to studies conducted by Attune Health.

BENEFICIAL OVER THE ALTERNATIVE

Opioids are the most commonly used method around the world to treat pain. They are prescribed by physicians worldwide but carry a host of negative implications. Firstly, they can be immensely addictive leading to opioid dependency in patients, especially ones suffering from chronic pain or pain that is difficult to diagnose. Secondly, the number of deaths due to opioid overdose is unimaginably high across the world and is consistently rising. Additionally, they can have



Opioids dependency threatens the lives of countless patients across the world.

plenty of side effects such as nausea, constipation, cognitive degeneration and respiratory ailments.

The consequences of long-term usage of opioids far supersede those of VR, since the farthest the latter can go is causing dizziness and discomfort. Brennan Spiegel, a doctor, opines that patients seeking pain relief will not become completely dependent on VR treatments as they do with opioids. However, VR usage in the medical field will not completely solve the opioid crisis but it will help to make new patients less dependent on medication in the beginning. They can be used in conjunction with opioids, used in lesser dosages, to help patients manage their pain as well as stress. In a study consisting of 18 patients, it was found that 39 per cent patients going through wound-care procedures requested lesser opioid dosages after receiving VR treatment for their pain.

APPLICATIONS OF VR IN THE FIELD

Virtual reality in the medical field has garnered a host of applications in the short time that they have been around. VR therapy can essentially be used to treat patients suffering from pain across all age groups and sporting any kind of pain. However, there are a few applications that have stood out and where VR treatment is extensively studied or used.

This technology is used to treat phantom limb pain regularly since VR can actually make them experience moving around painlessly. It is also used extensively for chronic pain since they reduce the need for opioids

as discussed above. The most common applications include the easing of temporary pain that some medical procedures cause such as dressing and stitching of wounds.

VR treatment is also commonly used to treat children since they have shown reductions in both acute and chronic pain, and also occupy their mind which is comparatively more fragile and frivolous than adults. Children often refuse medical care such as injections and other pain-inducing treatments in anticipation of the pain. In fact, doctors have to undertake special education to understand child behaviour and how to treat them better. VR therapy is a good negotiator since they get to be entertained while going through medical care. A study in



VR treatment is especially useful for treating children.

California consisting of 244 children from the ages of 2 to 16 showed that using VR for about 30 seconds before a vaccination for seasonal influenza resulted in a 45 to 74 per cent decrease in pain compared to the usual care methods. If something can make things less painful for children, we should do all we can to make it widespread, right?

VR treatment is also used during childbirth to lessen the intensity of the hours of contractions that precede the actual birth-giving process. This treatment allows many mothers to give birth without the assistance of drugs. So, apart from helping children, it can help expecting mothers as well. Maybe someday it can help us get rid of pain entirely from therapy. 